Ⓐ 2025-J118-1-EL-01/24

Pre-Junior Cycle Final Examination, 2025

Mathematics

Higher Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

DAY and MONTH of BIRTH	/ For example, 3rd February is entered as 03/02
NAME	
SCHOOL	
TEMPLATE	

School Stamp

For Examiner only	
Question	Mark
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	
11.	
12.	
Total	
%	

Ⓐ 2025-J118-1-EL-01/24

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:	

Question 1 **(Suggested maximum time: 5 minutes)**

Ten different numbers are listed in ascending order as follows:

a	7	10	11	b	19	29	c	45	54

(a) The 10 numbers have a **range** of 49.
Work out the value of a where $a \in \mathbb{N}$.

(b) The 10 numbers have a **median** of 16.
Work out the value of b where $b \in \mathbb{N}$.

(c) The 10 numbers have a **mean** of 23.
Work out the value of c where $c \in \mathbb{N}$.

Question 2 **(Suggested maximum time: 10 minutes)**

A local community centre runs a weekly fund-raising lottery.
To play this lottery game, you must pick any four numbers from the following natural numbers.

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24

(a) Describe in your own words what a natural number is.

(b) Jimmy picks the same four **square numbers** each week he plays.
Write down the four numbers Jimmy picks. One is done for you.

	4		

(c) Marion picks four numbers that are factors of 21.
Write down the four numbers Marion picks.

(d) Write down the probability that the first number drawn for the lottery next week will be a **prime number**.

(e) Jimmy says that he has a '1 in 24 chance' that one of his chosen numbers will be the first number drawn next week.
Marion says that she has 'over a 16% chance' that one of her numbers will be the first number drawn next week.

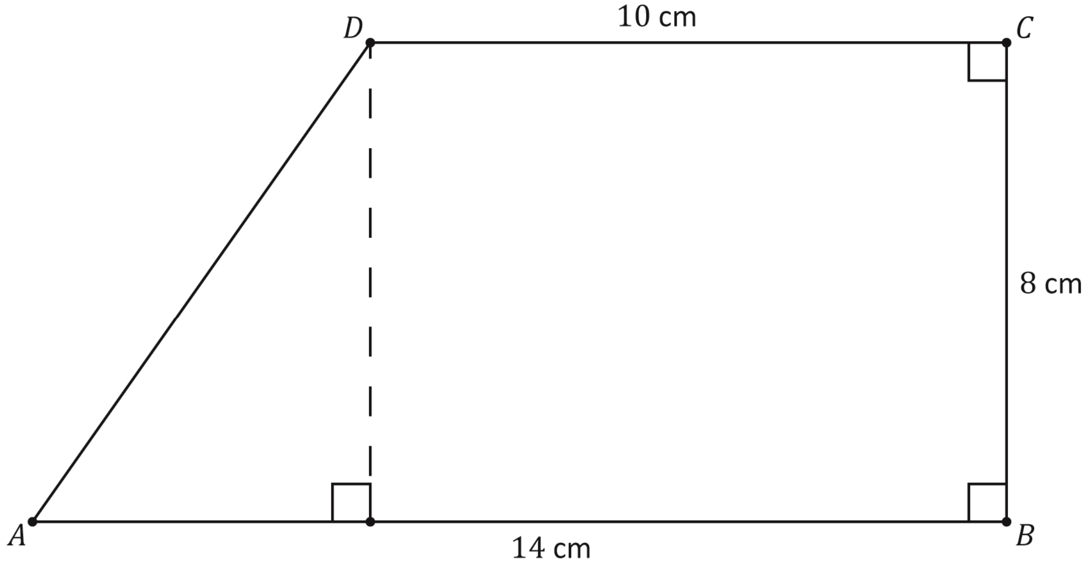
Tick (✓) the correct box to show who is correct.
Tick **one** box only. Justify your answer.

Jimmy correct	Marion correct	Both correct
☐	☐	☐

Justification:

Question 3 **(Suggested maximum time: 10 minutes)**

The diagram below shows a quadrilateral $ABCD$ (not to scale).
AB and DC are parallel and both lines are perpendicular to BC.
The lengths of some of the sides are given in the diagram.



(a) Tick (✓) the correct box to show the name this quadrilateral shape is commonly known as.
Tick **one** box only.

Rectangle	Parallelogram	Pentagon	Trapezium
☐	☐	☐	☐

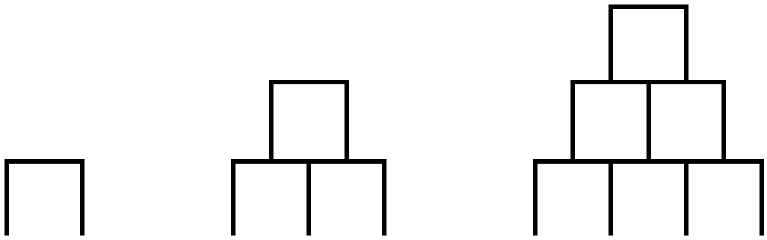
(b) Work out the area of the quadrilateral $ABCD$.

(c) Work out $|AD|$.
Give your answer in the form $a\sqrt{b}$, where $a, b \in \mathbb{N}$.

(d) Use trigonometry to work out $|\angle DAB|$
Give your answer in degrees, correct to 1 decimal place.

Question 4 **(Suggested maximum time: 10 minutes)**

The first three patterns in a sequence are shown below.
Each pattern consists of 1 cm upright and horizontal bars



Pattern 1 Pattern 2 Pattern 3

(a) Draw Pattern 4 in the sequence.

(b) Complete the table to show the total number of bars in each pattern in the sequence.

Pattern	1	2	3	4	5
Number of bars	3	8			

(c) Tick (✓) the correct box to show what kind of sequence is made by the number of bars in each pattern.
Tick **one** box only. **Justify** your answer.

linear ☐ quadratic ☐ exponential ☐

Justification:

(d) Show that the nth term of this sequence is $n^2 + 2n$.

(e) A pattern in this sequence is made using 99 bars.
Work out what pattern in the sequence this is.

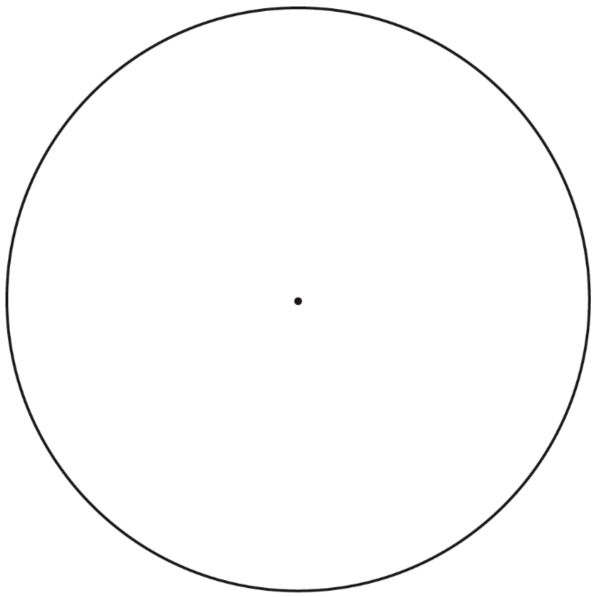
Question 5 **(Suggested maximum time: 10 minutes)**

Under the new Re-turn scheme, 1800 bottles and cans were recycled at a supermarket during the months of March, April and May.
The table below shows the number of bottles and cans recycled each month.

Month	March	April	May
Number of recycled items	400	635	

(a) Work out the number of items recycled at the supermarket during the month of May.

(b) Complete the **pie chart below**, to summarise the data above, showing the number of items recycled during the months of March, April and May.
Label each sector **and** the size of the angle clearly.
Show any working out and construction lines.



(c) A 500 ml soft drink costs €$2{\cdot}25$ which includes the Re-turn deposit of 15 cent.
Work out the Re-turn deposit as a **percentage** of the total cost of the soft drink.
Give your answer correct to the nearest precent.

(d) The Re-turn deposit of 25 cent represents $6{\cdot}25\%$ of the total cost of a 2 litre soft drink.
Work out the total cost of the soft drink.

Question 6 **(Suggested maximum time: 10 minutes)**

(a) A function is defined as $f(x) = 6x - 5$, where $x \in \mathbb{R}$.

(i) Work out the value of $f(3)$.

(ii) Work out the value of x for which $f(x) = 55$.

(iii) Work out the values of x for which $f(x) = x^2$.

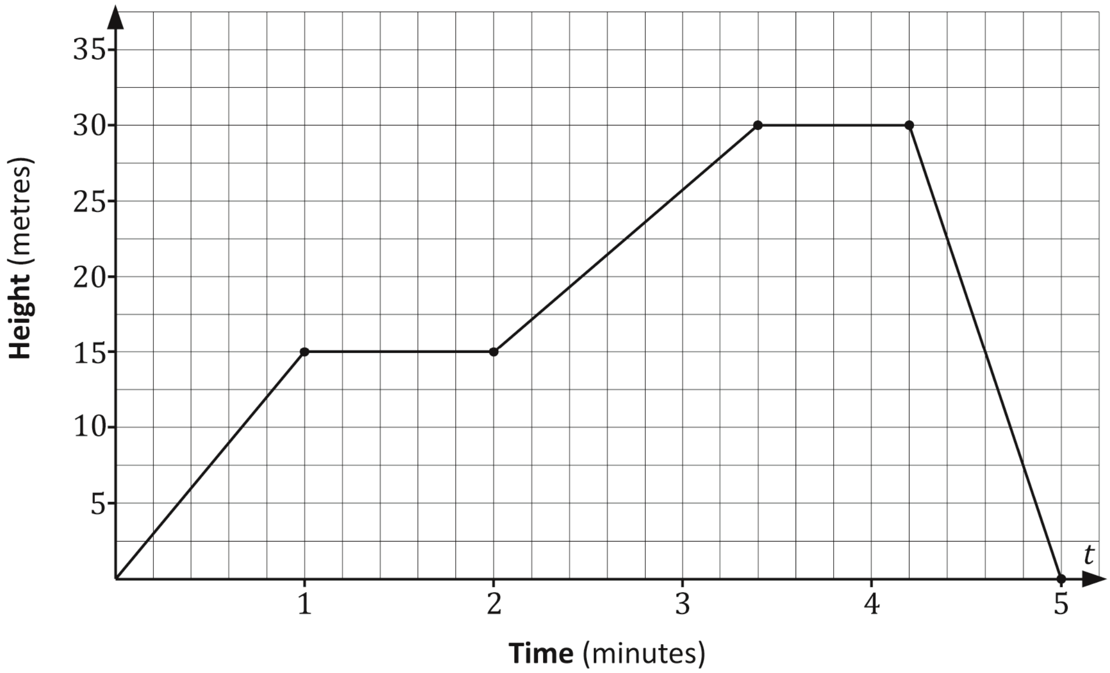
(b) Multiply out and simplify $(x+7)(x-2)$.

(c) Factorise and simplify $\frac{3x+6}{x^2-4}$.

(d) Solve the equation $x^2 - 6x - 10 = 0$.
Give each answer correct to 2 decimal places.

Question 7 **(Suggested maximum time: 10 minutes)**

The graph displays the height of a drone above ground level after t minutes, during a test flight.



(a) Using the graph answer the following questions:

(i) Write down how long the drone was in the air? Answer =

(ii) The drone's ability to stop and hover in the air was tested during the flight. Indicate using the letters **A** and **B**, the sections of the graph where the drone was possibly hovering.

(iii) Work out for how long the drone flew above the height of 15 metres. Give your answer in minutes and seconds.

minutes	seconds

(b) A second drone being tested ascended 25 metres at a constant speed for one minute. It hovered for 30 seconds before descending at a slower constant speed, reaching ground level after another two minutes.

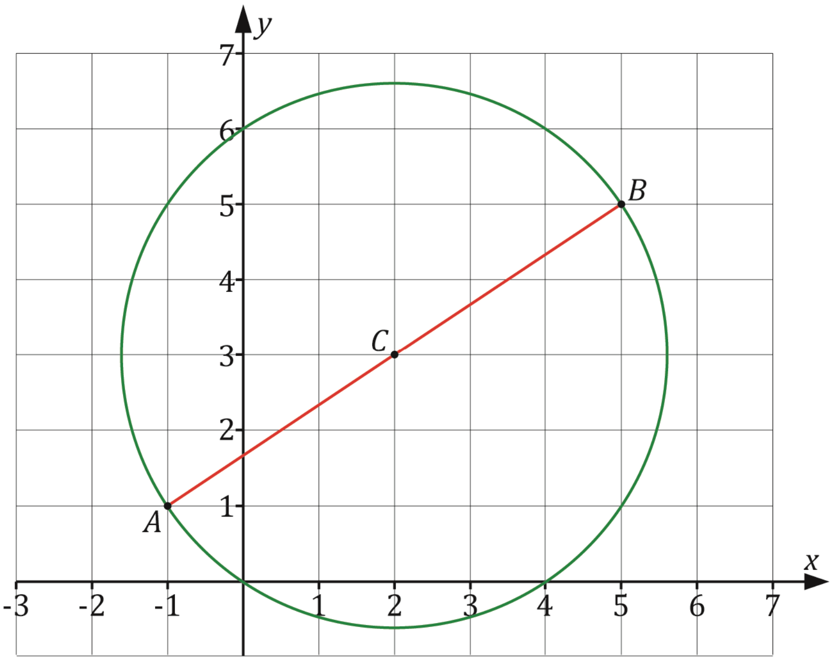
(i) On the given diagram, **draw** a graph to show the height of the second drone above ground level after t minutes.
Use the same axis and scales.

(ii) Work out the average speed of this drone during its descent.
Give your answer in metres per second, correct to 1 decimal place.

(c) The blades on each rotator can do 150 full rotations every second at this speed.
Work out how many rotations will it do in 1 hour at this same speed.
Give your answer in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{N}$.

Question 8 **(Suggested maximum time: 10 minutes)**

The co-ordinate diagram below shows a circle with the line segment $[AB]$ as it's diameter and the point $C(2, 3)$ as it's centre.



(a) Write down the co-ordinates of the point A and the point B.

$A =$ (,) $B =$ (,)

(b) Show that the point C is the midpoint of the line segment $[AB]$.

(c) Show that $|CB| = \sqrt{13}$.

(d) Work out the length of the circumference of the circle in the co-ordinate diagram.
Give your answer, correct to 1 decimal place.

(e) Another line segment $[PQ]$ is also a diameter of the circle and passes through the point C.
The equation of the line PQ is $3x + 2y = 12$.

(i) Show that AB is **perpendicular** to PQ.

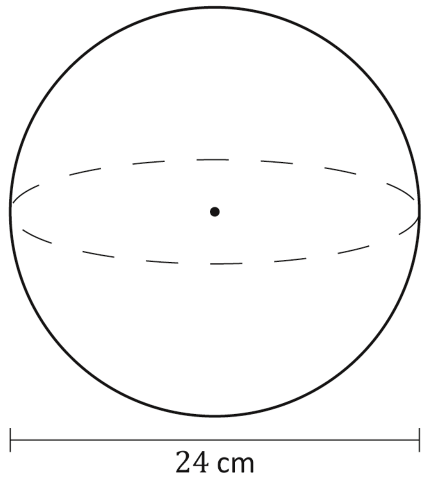
(ii) Construct the line segment $[PQ]$ in the co-ordinate diagram.

(iii) Write down the co-ordinates of the point P and the point Q.

$P =$ (,) $Q =$ (,)

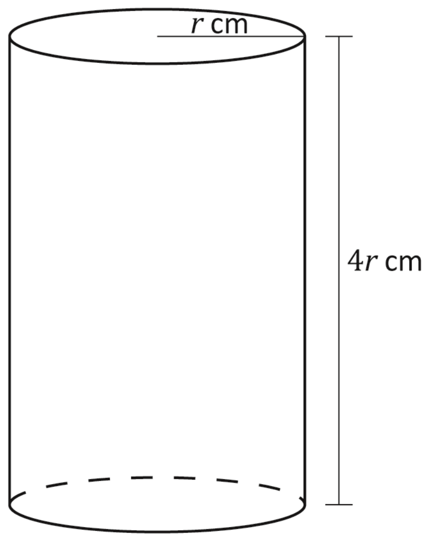
Question 9 **(Suggested maximum time: 5 minutes)**

A sphere has a diameter of 24 cm as shown.



(a) Find the volume of the sphere.
Give your answer in cm^3, in terms of π.

(b) A cylinder has a of radius r cm and a height $4r$ cm.



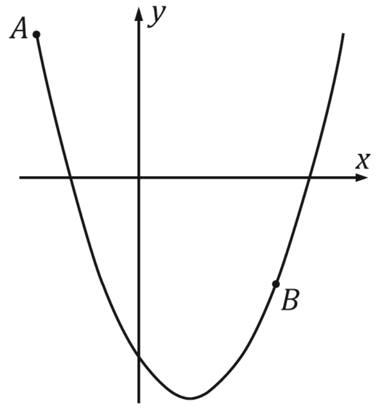
(i) Show that the volume of the cylinder is $4\pi r^3$.

(ii) The volume of the cylinder is equal to the volume of the sphere.
Work out the height of the cylinder.
Give your answer in cm, correct to 1 decimal place.

Question 10 **(Suggested maximum time: 10 minutes)**

The graph of the function $y = x^2 + px + q$
is shown where $p,\ q \in \mathbb{N}$ and where $x,\ y \in \mathbb{R}$.

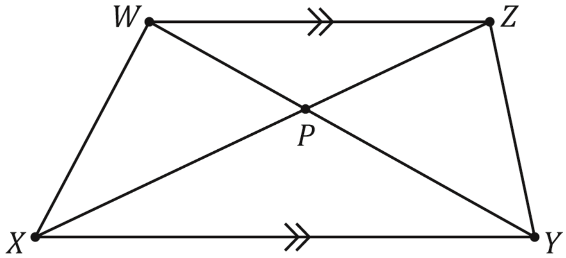
$A(-3,\ 8)$ and $B(4,\ -6)$ are two points of the function.



Work out the value of p and the value of q.

Question 11 **(Suggested maximum time: 10 minutes)**

$WXYZ$ is a quadrilateral called a trapezium.
All four sides of the trapezium are different lengths.
WZ is parallel to XY and the diagonals
WY and XZ intersect at the point P.



(a) For each of the following statements,
tick (✓) the correct box to show whether the statement is true or false.
In each case, tick **one** box only. Justify your answers.

(i) $|\angle XPY| = |\angle WPZ|$.

True ☐ False ☐

Justification:

(ii) Triangles ΔXYP and ΔWPZ are equiangular.

True ☐ False ☐

Justification:

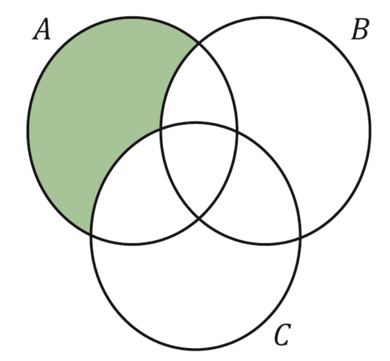
(iii) Area of $\Delta WXZ =$ area of ΔWYZ.

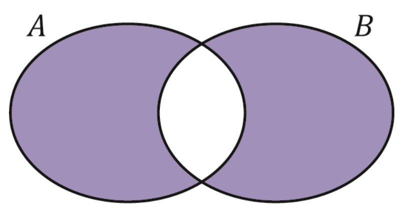
True ☐ False ☐

Justification:

Question 12 **(Suggested maximum time: 5 minutes)**

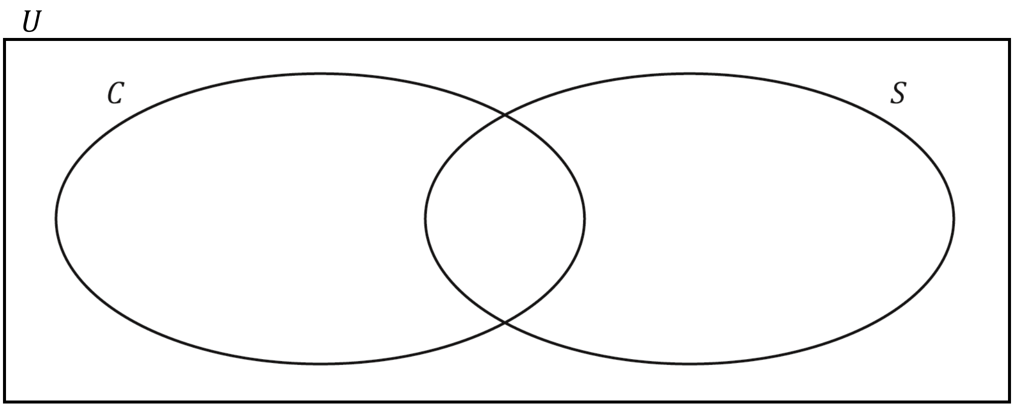
(a) A, B and C are three sets.
Complete the tables below by using set notation to describe the set illustrated by the shaded region in each of the Venn diagrams.

Venn diagram	
Set	

Venn diagram	
Set	

(b) A book shop has set up two stand displays of cookery books (C) and sports books (S).
On a particular day when 120 customers visited the shop,
22 customers bought at least one cookery book and
19 customers bought at least one sport book from the two display stands.
70% of the customers who visited the book shop did not buy a book from either stand.

Work out $\#(C \cap S)$, the number of customers who bought at least one book from each display stand. You may use the Venn diagram below to help answer the question.



$\#(C \cap S) =$

Page for extra work.
Label any extra work clearly with the question number and part.

Q. & part

Page for extra work.
Label any extra work clearly with the question number and part.

Q. & part	

Ⓐ

J.118

Do not write on this page

Pre-Junior Cycle Final Examination, 2025

Mathematics – Higher Level

Time: 2 hours

4e6b4069-6072-49fb-a557-f5acf7e96109